

PROJECT OVERVIEW

Smart UI Validator

From design intent to verified UI.



Minimal promise

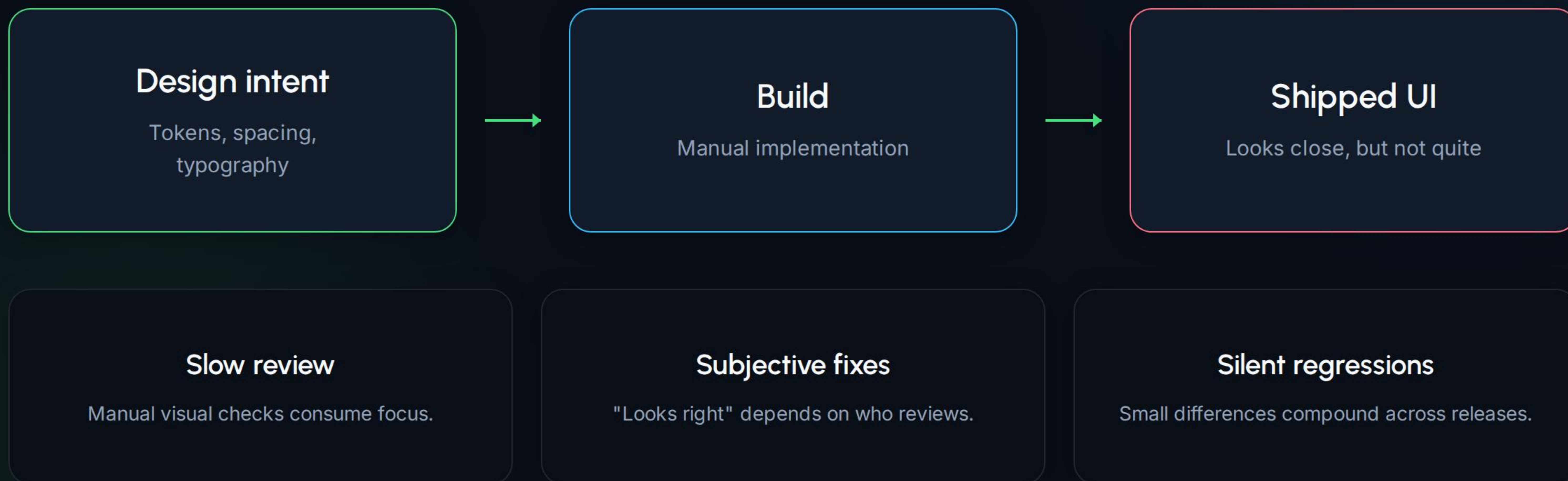
AI can help build UI. Smart UI makes the result measurable, reviewable, and safe.

A deterministic UI engineering engine for modern teams.

THE CHALLENGE

Designs drift during implementation

Precise designs pass through manual implementation, subjective review, and release pressure.



The gap between design and code is still reviewed manually more often than it is measured.

What if the feedback loop were automatic?

Every UI change could be rendered, compared, explained, and improved before manual review.



From subjective review

Every change can be rendered, compared, scored, and explained.

To repeatable evidence

Developers get clearer feedback. Designers get traceable alignment.

The shift: from subjective review to a repeatable verification loop.

What is Smart UI Validator?

A host-neutral toolchain that accepts design evidence, renders real code, measures drift, and supports safe repair.

Design evidence

Figma exports, SVG, PNG, structured context, component references.

Real browser

Runs the target UI in isolated Chromium using Playwright.

Safe repairs

Bounded changes, reviewable records, rollback when evidence regresses.

AI can propose changes; Smart UI measures what actually changed.

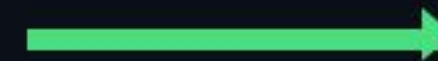
CORE IDEA

AI builds. Smart UI verifies.

The authoring side and verification side are different jobs.

Agent / Developer

- Interprets design intent
- Writes or repairs code
- Chooses trade-offs
- Asks for guidance when needed



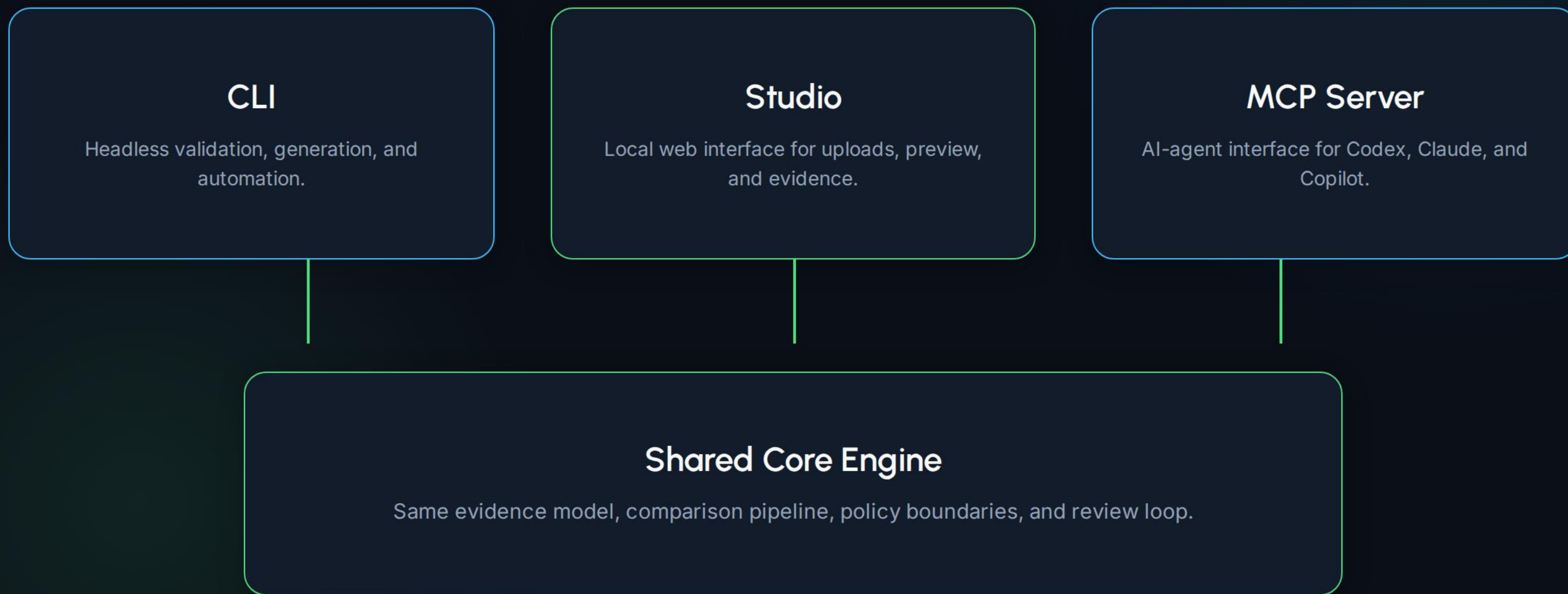
Smart UI

- Renders the implementation
- Measures visual and structural gaps
- Creates evidence
- Controls acceptance boundaries

Don't ask the agent whether it succeeded. Measure it.

One engine, three surfaces

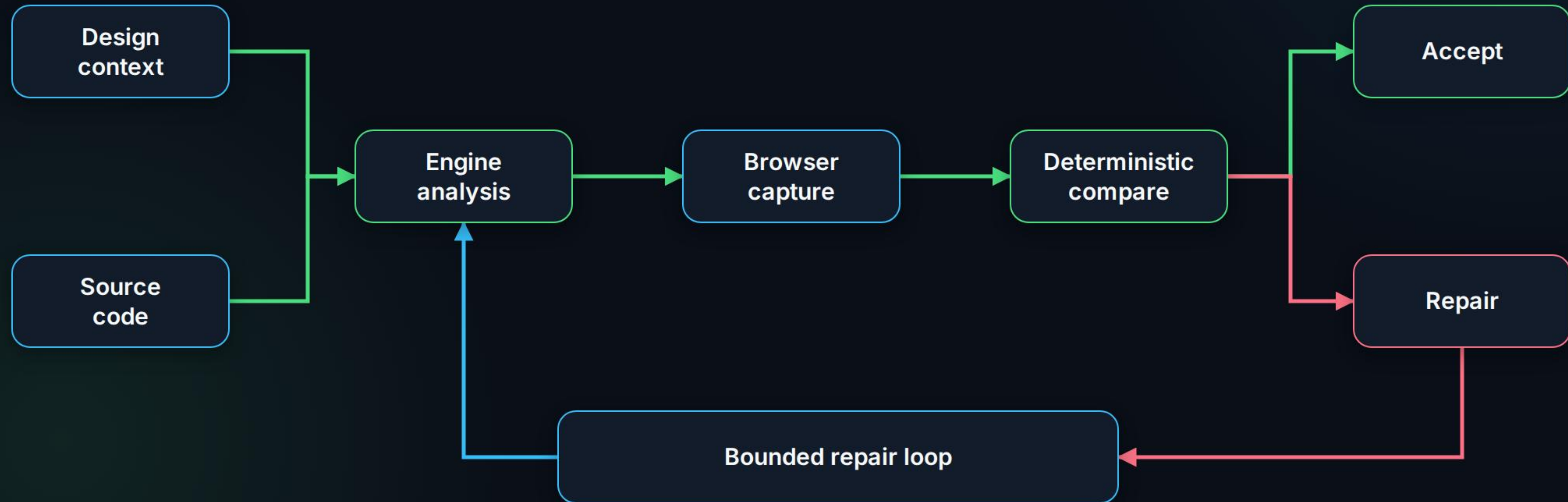
CLI, Studio, and MCP expose the same underlying validation engine.



Different entry points. One shared validation and evidence layer.

Render. Compare. Improve. Accept.

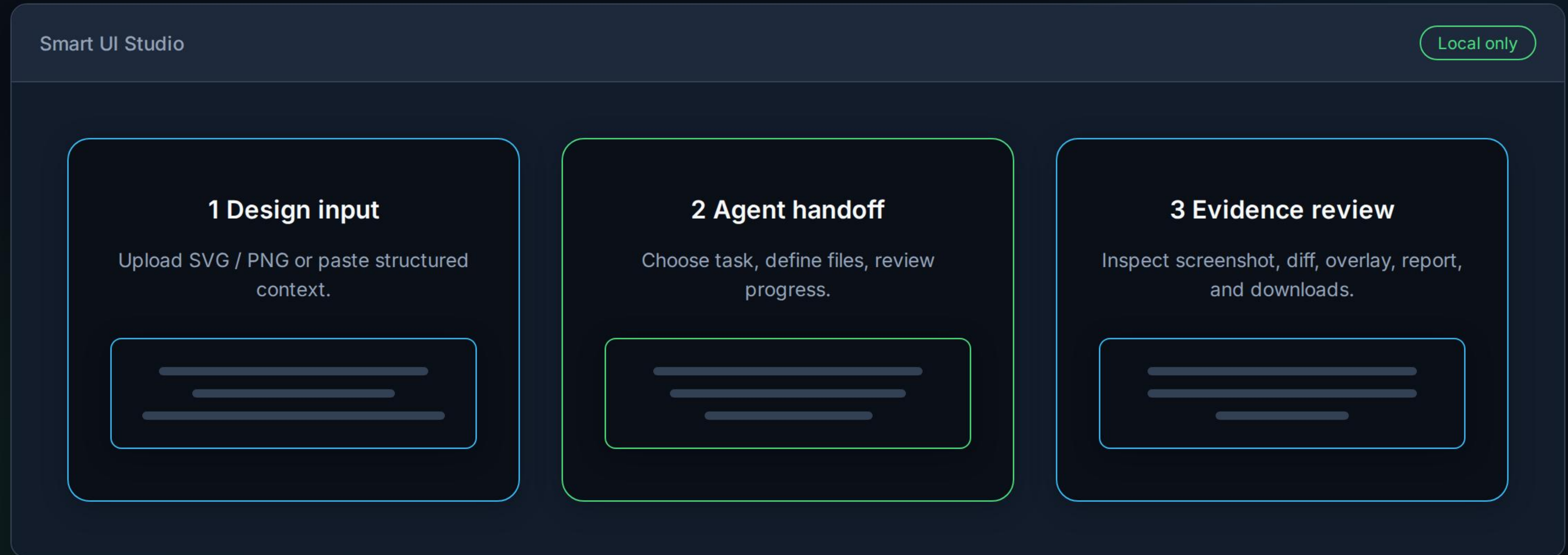
A simple loop: combine design context and source code, render in a browser, compare, then either accept or repair.



Every repair returns to the same deterministic compare step before acceptance.

A local Studio for guided review

The Studio gives teams a visual interface to the same core workflow.



Studio is the visual interface; the core engine remains the source of truth.

What does Smart UI measure?

It goes beyond pixels: layout, visuals, behavior, accessibility, evidence, and control boundaries.

Layout

spacing · sizing
alignment · overflow

Visuals

typography · color
borders · shadows

Behavior

viewports · states
runtime signals

Evidence

reference · diff
overlay · report

Accessibility

semantic structure
focus and contrast

Boundaries

allowed files
commands · rollback

Pixel-precise does not mean pixel-only.

Autonomy with guardrails

Smart UI keeps automation bounded, reviewable, and safe.

Exact write scope

Only approved files can be changed.

Command policy

Allowed checks are explicit and bounded.

Isolated browser

Capture happens inside a contained environment.

Rollback

Rejected repairs do not persist.

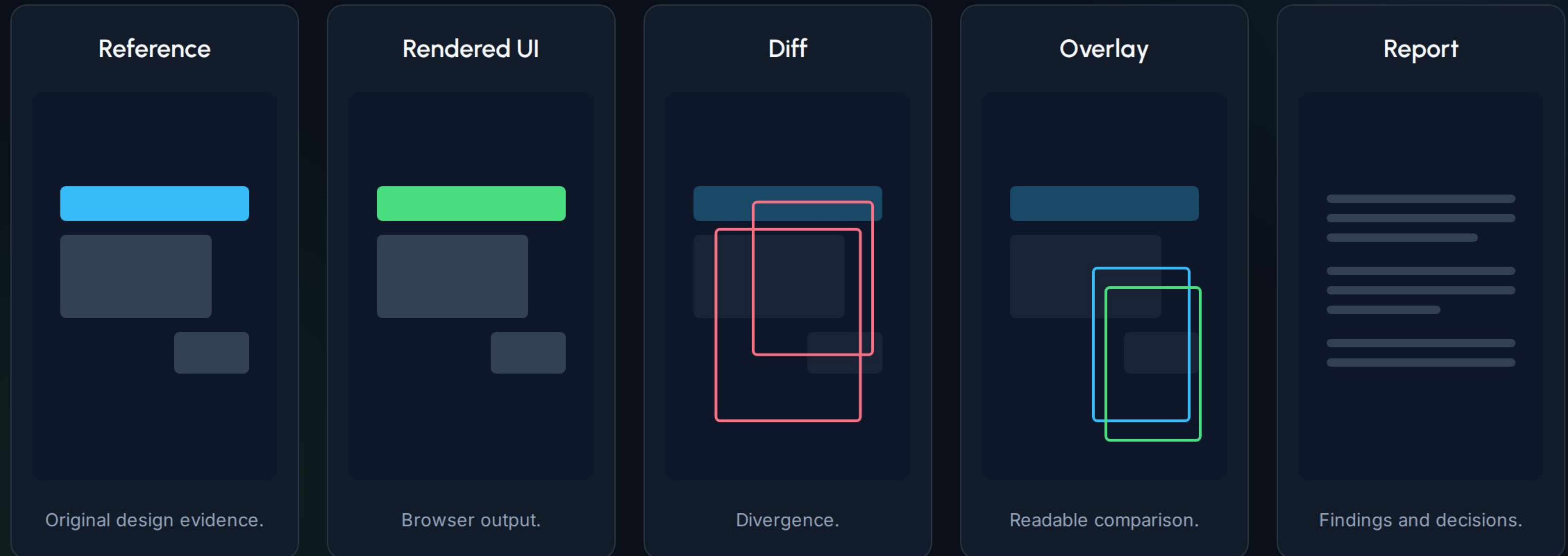
The agent can be creative inside a box. Smart UI defines the box.

Automation stays useful because permissions, checks, and rollback are explicit.

EVIDENCE, NOT OPINION

The output is reviewable

Teams see the same artifacts: design, rendered result, diff, overlay, and report.



Teams can discuss the same evidence and accept changes explicitly.

Where this goes next

The next steps complete the design-to-code loop — from cleaner handoff to autonomous delivery.



The developer moves from manual operator to guided reviewer and decision-maker.

Questions?

Thank you.